Georgii Bychkov

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ABOUT ME

Master's student at the Skolkovo Institute of Science and Technology specializing in Materials Science. My research interests lie at the intersection of computational physics, condensed matter theory, and machine learning. Currently, I am developing a machine learning framework to predict the catalytic activity of nanoparticles. Simultaneously, I am investigating the structural and electronic properties of Molybdenum-Sulfur nanoclusters using Density Functional Theory and custom Python-based algorithms for adsorption site analysis. I am dedicated to advancing digital materials science and am seeking opportunities to contribute to cutting-edge research in this field.

AREAS OF INTEREST

- ✓ Advancing digital materials science methodologies
- ✓ Applying machine learning techniques to materials science and physics problems
- ✓ Investigating the physics of nanoclusters and nanoparticles for applications in catalysis
- ✓ Developing computer models in condensed matter physics
- ✓ Utilizing microkinetic theory to predict catalytic activity

WORK EXPERIENCE

Research Engineer 2023 – present
Computer Design Laboratory, Center for Photonics and Two-Dimensional Materials, Moscow Institute of Physics and Technology, Russia

- ✓ *Supervisor*: Ivan Kruglov
- ✓ Investigating Molybdenum-Sulfur nanoclusters to evaluate and optimize their catalytic properties for water electrolysis and green hydrogen production.
- ✓ Developed an automated high-throughput Python pipeline to efficiently identify optimal molecular adsorption sites, significantly accelerating the computational screening process.
- ✓ Executing advanced Density Functional Theory (DFT) calculations and utilizing machine learning potentials to bridge *ab initio* accuracy with large-scale thermodynamic modeling.

TEACHING AND ORGANIZATIONAL EXPERIENCE

Organizer of Educational Astronomy Camps 2023 – present

- ✓ Developing and implementing intensive educational programs aimed at preparing middle school students for the regional stage of the All-Russian Olympiad.

Coach of the Moscow Region Astronomy Team 2021 – present

- ✓ Conducting lectures and seminars for middle and high school students to prepare them for national-level competitions.
- ✓ Mentoring students who subsequently achieved prize-winner and winner statuses at the regional and final stages of the All-Russian Olympiad.

SKILLS

Computational

- **Programming & Machine Learning**:
- ✓ **Python, ML & Parallel Computing**: Demonstrating high-level proficiency in Machine Learning, including engineering Graph Neural Networks (GNNs) and applying Natural Language Processing (NLP) specifically for materials science. Building robust models utilizing PyTorch, Scikit-learn, and CatBoost; automating workflows using ASE, Pymatgen, and Matminer, and accelerating computational processes via parallelization utilizing MPI (Message Passing Interface).
- ✓ **C++**: Utilizing academic-level knowledge to implement numerical methods and computational routines leveraging Boost and FenicsX.
- **Tools and Technologies**:
- ✓ **System & Deployment**: Utilizing Linux via Secure Shell for high-performance cluster computing, managing version control with Git, and configuring project builds with CMake.
- **Specialized Software for Materials Science**:
- ✓ **VASP**: Executing ab initio Density Functional Theory calculations and analyzing phonon dispersions via Phonopy.
- ✓ **JDFT-x**: Performing advanced Density Functional Theory calculations and running grand canonical simulations for electrochemistry.
- ✓ **USPEX**: Predicting stable crystal and nanocluster structures employing evolutionary algorithms.
- ✓ **LAMMPS & MLIP-3**: Running extensive molecular dynamics simulations leveraging machine-learning potentials.

Languages

- ✓ **English**: Attained B2 proficiency (Duolingo English Test Certificate, 2025).
- ✓ **Russian**: Speaking natively.

EDUCATION

Master of Science in Materials Science 2025 – present

Skolkovo Institute of Science and Technology, Skolkovo, Russia

- ✓ *Supervisor*: Alexander Kvashnin
- ✓ Focusing on the application of machine learning techniques to discover and model novel catalytic materials.

Bachelor of Science in Physics 2021 – 2025

Moscow Institute of Physics and Technology, Dolgoprudny, Russia

- ✓ Graduated with a Grade Point Average of 4.8 out of 5.0.
- ✓ Majored in General and Applied Physics within the Department of Physics and Technology of Nanostructures.
- ✓ Defended thesis: "Searching and studying nanocluster structures in the Molybdenum-Sulfur system using the USPEX evolutionary algorithm."

PUBLICATIONS

- ✓ Published research: Bychkov G. D., Kruglov I., Kravtsov K. V. (2025). A computational roadmap to stable Molybdenum-Sulfur nanoclusters: atomic structures and magic compositions. *Nanoscale*. DOI: 10.1039/D5NR02304A

CONFERENCES AND WORKSHOPS

- ✓ Presented at the 3rd Asia-Europe Symposium on Chemistry and Materials, Urumqi, China (August 20–22, 2025).
- ✓ Presented at the 67th All-Russian Scientific Conference of the Moscow Institute of Physics and Technology, Russia (March 31 – April 5, 2025).
- ✓ Delivered a poster presentation at "Matter and Materials", Skolkovo Institute of Science and Technology, Russia (March 3–4, 2025).
- ✓ Participated in the workshop "Computational Methods in Materials Science and Chemistry", Novgorod State University, Russia (November 23–26, 2024).
- ✓ Attended the Foresight Session: First All-Russian Conference on Computer Materials Science, Skolkovo Institute of Science and Technology, Russia (October 30 – November 2, 2023).

HOBBIES

- ✓ Running a personal channel dedicated to photography and astrophotography.
- ✓ Practicing the craft of wood burning.
- ✓ Engaging in physical activities, including weightlifting, dancing, basketball, and swimming.
- ✓ Exploring new cultures and environments through traveling.